

EDUCATION

- M.Sc., Computational Statistics and Machine Learning**, University College London, London, UK Sep 2024 — Sep 2025
- Thesis: Self-Causalizing LLM Inference for Robust Reasoning
 - Courses: Probabilistic and Unsupervised Learning, Supervised Learning, Bayesian Deep Learning, Statistical Model and Analysis, Open-Endedness and General Intelligence, Statistical NLP
- M.Sc., Computer Science**, National Taiwan University, Taipei, Taiwan Sep 2022 — Jun 2024
- GPA: 4.17/4.3
 - Thesis: Enhancing Retrieval Augmented Generation with Passage Combination (Paper: [📄](#)) (Code: [📁](#))
 - Relevant Courses: Statistical Foundation of Data Science, Machine learning, Applied Deep Learning, Big Data Systems, Computer Vision Practice with Deep Learning
- B.Sc., Undergraduate Honors Program of Electrical Engineering and Computer Science** Sep 2018 - Aug 2022
- National Yang Ming Chiao Tung University (Prev. National Chiao Tung University), Hsinchu, Taiwan
- Overall GPA: 3.93/4.3

RESEARCH AND WORK EXPERIENCE

- MediaTek Research UK** Jun 2024 — Aug 2024
- Deep Learning Intern
- Investigated transformer layer pruning and adaptive inference to reduce compute cost.
 - Explored diffusion-inspired methods for improving transformer representations.
 - Designed and evaluated experiments on reasoning benchmarks.
- Global Media Team, Yahoo!** Sep 2023 — Dec 2023
- Project Intern (Code: [📁](#))
- Developed and evaluated multitask learning methods for text readability assessment using unsupervised learning-to-rank with multiple readability signals.
 - Conducted statistical analysis and feature exploration (e.g. humor, subjectivity) to support reproducibility and further research.
- Global Media Team, Yahoo!** Jul 2023 — Aug 2023
- Research Engineering Intern
- Built a retrieval-augmented LLM chatbot for Yahoo news, integrating vector search, prompt optimization, and model quantization.
 - Enhanced answer quality, speed, and explainability through methods like max marginal relevance, asymmetric embeddings, and explainable RAG techniques.
- Center for Institutional Research & Data Analytics @ National Yang Ming Chiao Tung University** Sep — Dec 2020
- Part-time Data Analyst
- Statistically analyzed and visualized course selection data from different student groups, highlighting key trends and patterns, enabling stakeholders to make data-driven decisions.
- Institute of Information Science @ Academia Sinica, Taiwan** Jul — Aug 2020
- Summer Intern
- Researched efficient real-time processing for large-scale sensor networks.
 - Optimized multithreaded data collection from 7K+ PM2.5 sensors, cutting memory use by 75%.
 - Improved feature engineering, logging, and documented system architecture with UML.

PUBLICATIONS AND PREPRINTS

- **Ting-Wen Ko**, Jyun-Yu Jiang, Pu-Jen Cheng, "Beyond Independent Passages: Adaptive Passage Combination Retrieval for Retrieval Augmented Open-Domain Question Answering." Preprint, 2025. [📄](#)
- Yeoktatt Cheah*, Benjamin Cook*, Labeebah Islaam*, **Ting-Wen Ko***, Passawis Chaiyapattanaporn*, Zheng Jay Choy*, Noah Y. Siegel, "RL Training as a Tool: Understanding and Taxonomizing Faithfulness Metrics in LLMs." Under review, 2025. [📄](#)
- Alexander Inch*, Passawis Chaiyapattanaporn*, **Ting-Wen Ko***, Yuan Lu*, Yuchen Zhu*, Davide Paglieri, "BANDITPROMPT: Steering Creativity in Image Generation via Prompt Evolution and Bandit Optimization." Under review, 2025. [📄](#)

SELECTED PROJECTS

Danceability Prediction (Python / Sklearn / Pandas / NumPy) (Report:  • Predicted danceability of 17K+ tracks using XGBoost, SVM, and logistic regression. • Boosted accuracy by 17.8% via TF-IDF and SVD feature transformations. • Ranked in top 17% with detailed analysis on optimal feature combinations.	May 2022 — Jun 2023
CHEGG: Chinese Emo Graffiti Generator (Python / Pytorch / Huggingface) (Report:  Video:  • Built a diffusion-based system for emotion-rich graffiti generation. • Created a text-to-mask module to control image editing regions from prompts.	May 2022 — Jun 2023
Course Purchase Prediction (Python / Pytorch / Huggingface) (Code:  • Predicted user purchases using regression, matrix factorization, and kNN on Hahow online course platform data.	Dec 2022 — Jan 2023
Question Answering (Python / Pytorch / Huggingface) (Code:  • Built QA pipeline with RoBERTa/MacBERT achieving EM 84.3, ranked 36/274 (top 13%) in a competitive leaderboard.	Nov 2022
Intent Classification / Slot Tagging (Python / Pytorch / Huggingface) (Code:  • Trained RNN-based models (LSTM, GRU) for intent and slot tasks, achieving 93.2% accuracy and ranked top 9% .	Oct 2022
Relevant Table Searcher (Python / SentenceTransformers / Streamlit)  • Built a Streamlit app to retrieve relevant tables from documents using sentence embeddings and cosine similarity.	Dec 2023
Rotten Fruit Image Recognizer (Python / Pytorch / OpenCV) (Code:  • Investigated robustness against salt-and-pepper noise in classification. • Used CNNs and SVMs to classify fruit images under varying noise and backgrounds.	Jun 2021

TEACHING EXPERIENCE

System Programming Teaching Assistant (C) National Taiwan University	Sep 2022 — Dec 2022
• Designed and implemented a simulation of a context switch system utilizing non-local jumps and signals in a class assignment. • Conducted TA sessions, providing guidance and support to 100+ students on academic coursework.	

SKILLS

Frameworks	PyTorch, HuggingFace, vLLM, Weights & Biases, Langchain, Scikit-Learn
Toolkits	Python (7y), Git, Linux, Streamlit, Pandas, NumPy, C/C++, SQL

LANGUAGES

English (Fluent)	TOEFL (97/120); TOEIC (975/990) ; Part-time employee at the office of foreign affairs for one year, answering questions from international students; Enrolled in 10+ courses taught in English.
German (Basic)	A2.1 level; Exchange student in RWTH Aachen Germany for six months.
Chinese (Fluent)	

SERVICE

Student Volunteer The 46th International ACM SIGIR Conference	Jul 2023
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AWARDS & HONORS

2024	Phi Tau Phi Scholastic Honor Society of the Republic of China Honorary Membership
2022	Ministry of Education (MOE) Overseas Exchange Student Financial Assistance Grant (US \$4.6K)

ACTIVITY

Cooperative AI Summer School	Jul 2025
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Last updated July 14, 2025